

An L^2 Einstein Gap for Nonnegatively Curved Four-Manifolds

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Abstract

We prove a universal, scale-invariant L^2 gap for the trace-free Ricci tensor on simply connected closed four-manifolds with nonnegative sectional curvature. If the trace-free Ricci energy is sufficiently small, the manifold is diffeomorphic to S^4 , $\mathbb{CP}^2$, or $S^2 \times S^2$. The proof uses the classification of positive-Einstein, nonnegatively curved four-manifolds supplied by the companion zero-plane rigidity theorem, stated explicitly as the classification premise of our result. No auxiliary curvature, volume, diameter, injectivity-radius, or Sobolev bound is required. The conclusion concerns the smooth manifold, not an isometry of the original metric.

1 Introduction

For a Riemannian four-manifold (M, g) , let

$$E_g = \text{Ric}_g - \frac{\text{Scal}_g}{4}g, \quad \mathcal{D}(g) = \int_M |E_g|_g^2 d\mu_g.$$

The tensor E_g vanishes precisely when g is Einstein, provided that M is connected. The energy $\mathcal{D}$ is unchanged by constant rescaling in dimension four. It is therefore natural to ask whether a classification of Einstein metrics remains valid at the level of smooth manifolds when $\mathcal{D}$ is small. We address this question under nonnegative sectional curvature.

We take the exact Einstein classification in [11, Corollary 1.2] as the following premise.

Classification Assumption 1.1. Every connected smooth closed Riemannian four-manifold (N, h) satisfying

$$\text{Ric}_h = 3h, \quad \sec_h \geq 0$$

has universal Riemannian cover isometric to one of the following: the round S^4 of sectional curvature 1; Fubini–Study $\mathbb{CP}^2$ normalized by $\text{Ric} = 3g$; or $S^2(1/\sqrt{3}) \times S^2(1/\sqrt{3})$ with its product metric.

This premise is used only for an exactly Einstein metric; no stability statement is assumed.

Theorem 1.2. *Assume Classification Assumption 1.1. There is a universal constant $\varepsilon_0 > 0$ with the following property. If (M, g) is a connected, simply connected, smooth closed Riemannian four-manifold with $\sec_g \geq 0$ and*

$$\int_M \left| \text{Ric}_g - \frac{\text{Scal}_g}{4}g \right|_g^2 d\mu_g < \varepsilon_0,$$

then M is diffeomorphic, without a prescribed orientation, to the standard smooth S^4 , $\mathbb{CP}^2$, or $S^2 \times S^2$.

There are no auxiliary bounds on curvature, volume, diameter, injectivity radius, Sobolev constants, or curvature derivatives. The conclusion is about the underlying smooth manifold, rather than the original metric. This distinction matters because small L^2 error does not itself give pointwise curvature control.

Context and methods

The exact Einstein problem has been studied under several additional hypotheses. Gursky and LeBrun obtained the Fubini–Study conclusion for nonnegative sectional curvature and a nonzero positive-definite intersection form [6]. Cao and Tran established sectional-pinching criteria [1], and Liu classified the T^2 -invariant case on closed simply connected four-manifolds [9]. Here the classification premise is supplied by *Zero-Plane Rigidity for Einstein Four-Manifolds* [11, Corollary 1.2]. Our problem is instead to transfer that exact classification to metrics with small critical Ricci error.

The relation between Ricci curvature, scalar curvature, and topology is especially strong in dimension four: the Chern–Gauss–Bonnet formula controls the Euler characteristic by the total squared curvature. The almost-Schur inequality of De Lellis and Topping [5] controls the variance of scalar curvature by the trace-free Ricci tensor under nonnegative Ricci curvature. We use its Bianchi–Bochner argument to choose a scale in which the Ricci tensor is close in L^2 to $3g$.

Integral versions of Myers’ diameter argument combine second variation with control of curvature along many geodesics. The segment inequality of Cheeger and Colding [3] and the integral-curvature diameter estimates of Petersen and Sprouse [13, 15] provide the relevant context. We give the short segment-averaging argument needed here, together with an elementary derivation of the initial Sobolev inequality. These estimates prevent collapse before any smoothing is performed.

Hamilton’s Ricci flow [7] supplies the regularization. Perelman’s entropy monotonicity [12] propagates a scalar-curvature-weighted Sobolev inequality. The passage from entropy to Sobolev inequalities along Ricci flow is developed by Ye and Zhang [17, 18]; we include the form needed here, with constants that are uniform even before a common existence interval is known. Shi’s derivative estimates [14] and Hamilton’s compactness theorem [8] then turn curvature control at positive times into smooth convergence.

Smoothing from small critical curvature concentration was developed, in particular, by Chan, Chen, and Lee [2]. The formulation used here isolates a globally small *high part* of the curvature. Consider a sequence (M_j, g_j) satisfying the geometric hypotheses of Theorem 1.2 with $\mathcal{D}(g_j) \rightarrow 0$. After normalization, nonnegative sectional curvature gives, for a fixed threshold K_0 ,

$$\left\| (|\operatorname{Rm}_{g_j}| - K_0)_+ \right\|_{L^2(g_j)} \rightarrow 0.$$

A moving threshold propagates this estimate. It yields both a uniform Sobolev inequality and noncollapse along each existing flow; a curvature-concentration contradiction then establishes a common lifespan. The same tail bound controls the evolving Einstein error. Thus the high-part formulation supplies both the regularization and the defect estimate needed for the gap argument.

A second point concerns sectional curvature. Its nonnegativity is not assumed to persist under the flow. Instead, we bound the integral of its negative part by Ct . The limit is an exactly Einstein shrinking flow, and its explicit scaling converts this weak defect estimate into nonnegative sectional

curvature of the fixed Einstein metric. This gives the classification-independent compactness statement of Proposition 5.1; classification then identifies the limiting manifold.

Proof organization

Section 2 normalizes the sequence and establishes the initial geometric and Sobolev bounds. Section 3 obtains a uniform smoothing interval by controlling the high-curvature part. Section 4 propagates the Einstein error and estimates the negative sectional curvature. Section 5 constructs the compact Einstein limit, identifies its sectional curvature, and proves Theorem 1.2.

All manifolds are smooth, and *closed* means compact without boundary. We use $R = \text{Scal}$ and the curvature convention in which $\text{Rm}(X, Y, X, Y)$ is the sectional curvature of an orthonormal pair. The Laplacian is $\Delta = \text{tr } \nabla^2$. Norms and integrals use the displayed metric, or the current metric along a flow, and are not volume-normalized. Constants denoted by C or c may change from line to line. Uniformity always means independence of the sequence index, after discarding finitely many terms.

2 Normalization and initial geometry

Throughout the proof let (M_j, g_j) be connected, simply connected, closed four-manifolds with

$$\sec_{g_j} \geq 0, \quad \|E_{g_j}\|_2 \rightarrow 0. \quad (2.1)$$

The aim of this section is to obtain uniform geometric bounds without assuming a curvature maximum bound. We suppress j when considering one metric in the sequence.

Proposition 2.1. *After constant rescaling, the metrics in (2.1) satisfy*

$$\frac{1}{V} \int_M R d\mu = 12, \quad \|R - 12\|_2 \leq 4 \|E\|_2, \quad \|\text{Ric} - 3g\|_2 \rightarrow 0, \quad (2.2)$$

where $V = \text{Vol}(M, g)$. There are uniform positive constants v_0, V_0, D_0 such that

$$v_0 \leq V \leq V_0, \quad \text{diam}(M, g) \leq D_0. \quad (2.3)$$

Proof. Simple connectivity implies orientability and $b_1 = 0$. Poincaré duality therefore gives $\chi(M) = 2 + b_2(M) \geq 2$. By Chern–Gauss–Bonnet,

$$2 \leq \chi(M) \leq C \int_M |\text{Rm}|^2 d\mu. \quad (2.4)$$

On the space of algebraic curvature tensors, the maximum absolute sectional curvature is a norm: sectional curvatures determine the tensor. Equivalence of norms in this finite-dimensional space, together with $0 \leq \sec \leq R/2$, gives a dimensional constant C_0 such that

$$|\text{Rm}| \leq C_0 R. \quad (2.5)$$

The average scalar curvature is positive, since otherwise $R \equiv 0$ and (2.5) contradicts (2.4). Rescale so that its average is 12. The L^2 norm of E is unchanged.

For completeness, the almost-Schur argument [5, Section 2.1] in the form needed here is as follows. Solve $\Delta f = R - 12$. The contracted Bianchi identity says $\operatorname{div} E = \frac{1}{4}dR$, so integration by parts gives

$$\int_M (R - 12)^2 d\mu = 4 \int_M \langle E, \nabla^2 f \rangle d\mu.$$

Since $\operatorname{Ric} \geq 0$, the integrated Bochner formula yields $\|\nabla^2 f\|_2 \leq \|\Delta f\|_2$. Cauchy–Schwarz proves the second assertion in (2.2). Orthogonality of the trace and trace-free parts gives

$$\|\operatorname{Ric} - 3g\|_2^2 = \|E\|_2^2 + \frac{1}{4} \|R - 12\|_2^2 \longrightarrow 0. \quad (2.6)$$

Moreover,

$$\int_M R^2 d\mu = 144V + \|R - 12\|_2^2.$$

Combining this identity with (2.4) and (2.5) gives $V \geq v_0 > 0$ for all sufficiently large j .

It remains to control the diameter. Write $D = \operatorname{diam}(M, g)$ and choose points realizing it. Let A_1, A_2 be the balls of radius $D/4$ about these points. Bishop–Gromov comparison gives $\operatorname{Vol}(A_i) \geq 8^{-4}V$, by comparison with the ball of radius $2D$. For any nonnegative continuous function H , the segment inequality is

$$\int_{A_1 \times A_2} \int_{\gamma_{xy}} H ds d\mu(x) d\mu(y) \leq CD(\operatorname{Vol}(A_1) + \operatorname{Vol}(A_2)) \int_M H d\mu, \quad (2.7)$$

where γ_{xy} is the minimizing geodesic, unique for almost every pair. This is the nonnegative-Ricci case of [3, Theorem 2.11]. To see the estimate directly, fix x and use polar coordinates $y = \exp_x(r\theta)$ before cut time. On the half-segment $r/2 \leq s \leq r$, Jacobian comparison gives

$$J_x(r, \theta) \leq (r/s)^3 J_x(s, \theta) \leq 8J_x(s, \theta).$$

Changing the order of integration bounds the integral over this half, and over y , by $CD \int_M H d\mu$. Integrating in $x \in A_1$ gives its contribution to (2.7). Reverse the geodesic and fix y to treat the other half. The cut locus has measure zero.

Apply (2.7) to $H = |\operatorname{Ric} - 3g|$. Some minimizing geodesic between the two balls has length $\ell \geq D/2$ and satisfies

$$\int_0^\ell H(\gamma(s)) ds \leq \delta_j D, \quad \delta_j = CV^{-1/2} \|\operatorname{Ric} - 3g\|_2 \longrightarrow 0. \quad (2.8)$$

Use three parallel orthonormal normal fields along γ , each multiplied by $\sin(\pi s/\ell)$, in the index form. Summing gives

$$\frac{3\pi^2}{2\ell} \geq \int_0^\ell \sin^2(\pi s/\ell) \operatorname{Ric}(\dot{\gamma}, \dot{\gamma}) ds \geq \frac{3\ell}{2} - \delta_j D.$$

Because $D \leq 2\ell$, we obtain $(3/2 - 2\delta_j)\ell^2 \leq 3\pi^2/2$. Thus D is uniformly bounded. Bishop comparison now gives the uniform upper volume bound as well. $\square$

2.1 A support estimate and the Sobolev inequality

We will need Sobolev inequalities both initially and along the flow. The following elementary truncation lemma allows us to derive both from estimates for functions with small support.

Lemma 2.2. *Let (M, g) be a closed four-manifold, $a > 0$, and $P \geq 0$ a smooth function. Set*

$$\mathcal{E}(f) = \int_M (a|\nabla f|^2 + Pf^2) d\mu.$$

Suppose that for every Lipschitz function f , with $m = \text{Vol}\{f \neq 0\}$, one has

$$\|f\|_2^2 \leq Cm^{1/2}\mathcal{E}(f). \quad (2.9)$$

Then $\|f\|_4^2 \leq C'\mathcal{E}(f)$ for every such f , where C' depends only on C .

Proof. For $k \in \mathbb{Z}$ define

$$m_k = \text{Vol}\{|f| > 2^k\}, \quad f_k = \min\{(|f| - 2^k)_+, 2^k\}.$$

The assumed estimate implies

$$2^{2k}m_{k+1} \leq \|f_k\|_2^2 \leq Cm_k^{1/2}\mathcal{E}(f_k).$$

The gradients of f_k occupy disjoint layers up to null sets, and $\sum_k f_k^2 \leq C_1 f^2$ pointwise by a geometric series. Consequently $\sum_k \mathcal{E}(f_k) \leq C_2 \mathcal{E}(f)$, with numerical constants. Put $b_k = 2^{4k}m_k$. The preceding estimate becomes

$$b_{k+1} \leq 16Cb_k^{1/2}\mathcal{E}(f_k).$$

The sum $B = \sum_k b_k$ is finite and comparable, with numerical constants, to $\int_M |f|^4 d\mu$: integrate the geometric series $\sum_{2^k < |f|} 2^{4k}$ pointwise. Summing over k and using $b_k^{1/2} \leq B^{1/2}$ gives $B \leq 16CC_2B^{1/2}\mathcal{E}(f)$. Division by $B^{1/2}$ when $B > 0$ proves the result; the case $B = 0$ is immediate. $\square$

Proposition 2.3. *The normalized metrics of Proposition 2.1 satisfy*

$$\|f\|_4^2 \leq S \int_M (|\nabla f|^2 + f^2) d\mu \quad (2.10)$$

for all Lipschitz functions f , with a uniform constant S .

Proof. We establish (2.9) with $a = P = 1$. Let $\bar{f} = V^{-1} \int_M f d\mu$. Polar integration of $|f(x) - f(y)|$ along minimizing segments yields

$$|f(x) - \bar{f}| \leq \frac{CD^4}{V} \int_M \frac{|\nabla f(y)|}{d(x, y)^3} d\mu(y). \quad (2.11)$$

Indeed, on a ray from x , the coefficient of $|\nabla f|$ at distance s is bounded using

$$\int_s^{\text{cut}(\theta)} J_x(r, \theta) dr \leq \frac{J_x(s, \theta)}{s^3} \int_s^D r^3 dr \leq \frac{D^4 J_x(s, \theta)}{4s^3}.$$

This proves (2.11); absolute continuity along almost every ray justifies it for Lipschitz functions.

For any measurable set A of volume $m > 0$, comparison with Euclidean polar measure gives

$$\sup_x \int_A d(x, y)^{-3} d\mu(y) \leq Cm^{1/4}.$$

To verify this, split the integral at radius $\rho = m^{1/4}$. The inner integral is at most $C\rho$ because $J_x(r, \theta) \leq r^3$; the outer integral is at most $m\rho^{-3}$. By symmetry the same bound holds with x and y interchanged. The integral operator with this kernel, restricted to A in both variables, therefore has L^2 norm at most $Cm^{1/4}$ by the Schur estimate.

Take $A = \{f \neq 0\}$. The gradient vanishes almost everywhere off A , and (2.3) bounds D^4/V . Taking $L^2(A)$ norms in (2.11) gives

$$\|f\|_2 \leq Cm^{1/4} \|\nabla f\|_2 + \sqrt{m} |\bar{f}| \leq Cm^{1/4} \|\nabla f\|_2 + \frac{m}{V} \|f\|_2.$$

If $m/V \leq 1/2$, absorb the last term and square. If $m/V > 1/2$, then $m \geq v_0/2$, and (2.9) follows directly from the $\int f^2$ term in the energy. Lemma 2.2 completes the proof. $\square$

3 A uniform smoothing interval

Let $g_j(t)$ be the maximal smooth unnormalized Ricci flows with the normalized initial metrics above:

$$\partial_t g_j = -2 \operatorname{Ric}_{g_j}, \quad g_j(0) = g_j.$$

Short-time existence and the curvature continuation criterion are standard on closed manifolds [7]. Initially each flow has its own maximal time $\mathcal{T}_j > 0$. Until a common lower bound for these times is proved, every estimate refers only to existing times. The scalar evolution and volume variation are

$$\partial_t R = \Delta R + 2|\operatorname{Ric}|^2, \quad \partial_t d\mu = -R d\mu. \quad (3.1)$$

Thus $R \geq 0$ and $\operatorname{Vol}(M, g_j(t)) \leq V_0$ throughout smooth existence. We first propagate a Sobolev inequality with a scalar-curvature weight, then use smallness of the high-curvature part to remove that weight.

3.1 Entropy supplies a weighted Sobolev inequality

Proposition 3.1. *There is a uniform S_1 such that, whenever $0 \leq t < \min\{1, \mathcal{T}_j\}$,*

$$\|f\|_4^2 \leq S_1 \int_M (4|\nabla f|^2 + (R+1)f^2) d\mu \quad (3.2)$$

for every Lipschitz function f on $(M_j, g_j(t))$.

Proof. In dimension four, Perelman's entropy is

$$\begin{aligned} \mu(g, \tau) = \inf_{\int f^2 d\mu = 1} \left\{ \tau \int_M (4|\nabla f|^2 + Rf^2) d\mu - \int_M f^2 \log f^2 d\mu \right. \\ \left. - 2 \log(4\pi\tau) - 4 \right\}, \end{aligned}$$

where the infimum is over smooth positive f . Its monotonicity gives

$$\mu(g_j(t), \tau) \geq \mu(g_j(0), \tau + t), \quad \tau > 0 \quad (3.3)$$

[12, Section 3]. Here $\tau + t$ is an entropy parameter, not a later flow time, so (3.3) imposes no additional lifespan requirement.

At time zero, Jensen's inequality and (2.10) give

$$\int_M f^2 \log f^2 d\mu \leq \log \int_M f^4 d\mu \leq 2 \log(S(y+1)), \quad y = \int_M |\nabla f|^2 d\mu,$$

for $\int f^2 d\mu = 1$. Since $R \geq 0$,

$$\mu(g_j(0), \tau) \geq \inf_{y \geq 0} \{4\tau y - 2 \log(\tau(y+1))\} - C \geq -C' \quad (0 < \tau \leq 2).$$

The last bound follows by writing $z = \tau(y+1)$: the expression in braces is $4z - 2 \log z - 4\tau$, uniformly bounded below in this range. Consequently (3.3) gives, at all times under consideration,

$$\int_M f^2 \log f^2 d\mu \leq \tau \mathcal{E}_0(f) - 2 \log \tau + C, \quad \mathcal{E}_0(f) = \int_M (4|\nabla f|^2 + Rf^2) d\mu, \quad (3.4)$$

whenever $\|f\|_2 = 1$ and $0 < \tau \leq 1$. Approximation of $|f|$ by positive smooth functions extends this to Lipschitz f , with $0 \log 0 = 0$.

Let $m = \text{Vol}\{f \neq 0\}$. For a normalized f , Jensen's inequality on its support gives $\int f^2 \log f^2 d\mu \geq \log(1/m)$. Choose $\tau = (1 + \mathcal{E}_0(f))^{-1}$ in (3.4). It follows that

$$\log(1/m) \leq 2 \log(1 + \mathcal{E}_0(f)) + C.$$

Exponentiation, followed by homogeneity for arbitrary f , yields

$$\|f\|_2^2 \leq C m^{1/2} (\mathcal{E}_0(f) + \|f\|_2^2).$$

Apply Lemma 2.2 with $a = 4$ and $P = R + 1$. □

3.2 The high-curvature part stays small

In orthonormal frames evolving by $\partial_t e_i = \text{Ric}^\sharp e_i$, the curvature equation has the form

$$\partial_t \text{Rm} = \Delta \text{Rm} + \mathcal{Q}(\text{Rm}), \quad |\mathcal{Q}(\text{Rm})| \leq C_q |\text{Rm}|^2. \quad (3.5)$$

Here time derivatives are derivatives of the moving-frame components; the connection Laplacian has the sign fixed in Section 1. It follows that $u = |\text{Rm}|$ satisfies $(\partial_t - \Delta)u \leq C_q u^2$ wherever $u > 0$.

Choose the scalar function K by

$$K(0) = 12C_0, \quad K' = C_q K^2, \quad (3.6)$$

where C_0 is from (2.5). Fix $T \in (0, 1/6)$ so small that K is bounded on $[0, T]$, and set

$$w_j(t) = (|\text{Rm}_{g_j(t)}| - K(t))_+.$$

At time zero, (2.5) implies

$$0 \leq w_j(0) \leq C_0(R_{g_j} - 12)_+, \quad \|w_j(0)\|_2 \rightarrow 0. \quad (3.7)$$

Proposition 3.2. *After discarding finitely many indices, there are uniform constants S_2, C such that, for $0 \leq t < \min\{T, \mathcal{T}_j\}$,*

$$\|f\|_4^2 \leq S_2 \int_M (|\nabla f|^2 + f^2) d\mu, \quad (3.8)$$

$$\int_M w_j(t)^2 d\mu \leq e^{Ct} \int_M w_j(0)^2 d\mu_{g_j}, \quad (3.9)$$

$$\int_M |\text{Rm}_{g_j(t)}|^2 d\mu \leq C. \quad (3.10)$$

The Sobolev estimate holds for every Lipschitz function f . In particular, the supremum of $\|w_j(t)\|_2$ over these existing times tends to zero.

Proof. We first work on any interval where $\|w\|_2 \leq \eta$ for a small fixed $\eta > 0$. Since $R \leq C|\text{Rm}| \leq C(K + w)$, the last term in (3.2) is bounded by a fixed multiple of $\int f^2$ plus

$$C \int_M w f^2 d\mu \leq C\eta \|f\|_4^2.$$

Choose η so that this term is absorbed into the left side. This gives (3.8) with a fixed S_2 ; decreasing η further does not change the chosen S_2 .

On $\{u > K\}$, the equation for K removes the bounded-background reaction:

$$(\partial_t - \Delta)(u - K) \leq C_q(u^2 - K^2) = C_q(w^2 + 2Kw).$$

Integration by parts and (3.1) therefore give

$$\frac{d}{dt} \int_M w^2 d\mu \leq -2 \int_M |\nabla w|^2 d\mu + 2C_q \int_M w^3 d\mu + 4C_q K \int_M w^2 d\mu. \quad (3.11)$$

There is no adverse volume term, since $-\int R w^2 d\mu \leq 0$. The calculation is valid for the positive part: $K > 0$, so u is smooth near $\{u \geq K\}$; the function $(u - K)_+^2$ is continuously differentiable, and Lipschitz truncation justifies integration by parts.

Hölder's inequality and (3.8) give

$$\int_M w^3 d\mu \leq \eta \|w\|_4^2 \leq \eta S_2 \int_M (|\nabla w|^2 + w^2) d\mu.$$

Decrease η so that $2C_q \eta S_2 \leq 1$. Then $Y(t) = \int w^2 d\mu$ satisfies $Y' \leq CY$ as long as $Y^{1/2} \leq \eta$. By (3.7), for large j we have $Y(0)e^{CT} < \eta^2$. Gronwall's inequality and continuity exclude a first time with $Y^{1/2} = \eta$ before $\min\{T, \mathcal{T}_j\}$. This proves (3.8) and (3.9) on the entire existing interval. Finally, $u \leq K + w$ and $\text{Vol}(M, g_j(t)) \leq V_0$ give (3.10). $\square$

3.3 Noncollapse and the common lifespan

No common existence time has been used in this argument. We now use its two outputs, uniform Sobolev control and a vanishing curvature tail, to establish one.

Lemma 3.3. *There is a uniform $v_* > 0$ such that at every time covered by Proposition 3.2,*

$$\text{Vol}_{g_j(t)} B_{g_j(t)}(x, r) \geq v_* r^4 \quad \text{for all } x \in M_j, \quad 0 < r \leq 1. \quad (3.12)$$

Proof. Write $V_x(r) = \text{Vol } B(x, r)$ on one smooth slice. Insert into (3.8) the distance cutoff that is 1 on $B(x, r/2)$, vanishes outside $B(x, r)$, and has gradient at most $2/r$. Then

$$V_x(r/2)^{1/2} \leq Cr^{-2}V_x(r).$$

For $a(r) = V_x(r)/r^4$, this gives $a(r) \geq ca(r/2)^{1/2}$. Iteration yields

$$a(r) \geq c^{2(1-2^{-k})}a(2^{-k}r)^{2^{-k}}.$$

For this fixed smooth metric, $a(s) \rightarrow \omega_4$ as $s \downarrow 0$, where ω_4 is the volume of the Euclidean unit ball in $\mathbb{R}^4$. Letting $k \rightarrow \infty$ gives $a(r) \geq c^2$. No uniform scale for the Euclidean asymptotic is required. $\square$

Proposition 3.4. *For all sufficiently large j , the flow $g_j(t)$ exists on $[0, T]$. There is a uniform constant A such that*

$$\max_{M_j} |\text{Rm}_{g_j(t)}| \leq \frac{A}{t}, \quad 0 < t < T. \quad (3.13)$$

Proof. Fix $A > 2$ with $A/T \geq 1$ and $A/T \geq 4 \max_{[0, T]} K$. Suppose that $t \max_M u(t)$ first reaches A at an existing time $t < T$. Set $Q = \max_M u(t) = A/t$. On $[t - Q^{-1}, t]$ the curvature is at most $2Q$, because $t - Q^{-1} \geq t/2$ and this is a first crossing. Parabolic rescaling gives a smooth closed flow on $[0, 1]$ whose curvature norm is at most 2. Shi's first derivative estimate [14] gives

$$|\nabla \text{Rm}|(t) \leq CQ^{3/2}.$$

This estimate requires no initial derivative or injectivity-radius bound; one may use the local formulation in [10, Theorem 3.27] with the entire closed manifold as its domain.

Let x realize the curvature maximum. Choose a fixed $c \in (0, 1]$ small enough that on $B_{g_j(t)}(x, cQ^{-1/2})$ the derivative estimate implies $u \geq 3Q/4$. Since $Q \geq 4K(t)$, we have $w \geq Q/2$ there. Also $Q \geq 1$, so this radius is at most 1. Lemma 3.3 now gives

$$\int_M w(t)^2 d\mu \geq \frac{Q^2}{4} v_* c^4 Q^{-2} = \frac{v_* c^4}{4}.$$

This contradicts the uniform convergence of the tail norm to zero in Proposition 3.2. Thus, for large j , no crossing occurs, and (3.13) holds throughout existing times below T .

If $\mathcal{T}_j \leq T$, the same bound controls curvature on $[\mathcal{T}_j/2, \mathcal{T}_j]$. Curvature is already bounded on the earlier compact interval by smoothness of the individual flow. The curvature continuation criterion contradicts maximality. Hence $\mathcal{T}_j > T$. $\square$

4 The Einstein and sectional-curvature defects

The common interval now permits compactness at positive times. To identify the eventual limit, we must retain information from time zero: the Einstein error tends to zero, and the integral negative sectional curvature remains of order t .

Proposition 4.1. *Set $\lambda(t) = 3/(1 - 6t)$. Along the flows above,*

$$\sup_{0 \leq t < T} \left\| \text{Ric}_{g_j(t)} - \lambda(t)g_j(t) \right\|_{L^2(g_j(t))} \longrightarrow 0. \quad (4.1)$$

Proof. Write $L = \text{Ric} - \lambda g$. In moving orthonormal frames, define the curvature action on a symmetric tensor B by

$$(\text{Rm}(B))_{ij} = \sum_{k,l} R_{ikjl} B_{kl}; \quad \text{Rm}(g) = \text{Ric}.$$

The Ricci evolution in these frames is

$$\partial_t \text{Ric} = \Delta \text{Ric} + 2 \text{Rm}(\text{Ric}).$$

The frame motion cancels the -2Ric^2 term in the fixed-frame formula; the components of g in the moving frames are constant. Since $\lambda' = 2\lambda^2$, subtraction yields

$$\partial_t L = \Delta L + 2 \text{Rm}(L) + 2\lambda L. \quad (4.2)$$

Using $R \geq 0$, $|\text{Rm}| \leq K + w$, and boundedness of K and λ on $[0, T]$, we obtain

$$\frac{d}{dt} \int_M |L|^2 d\mu \leq -2 \int_M |\nabla L|^2 d\mu + C \int_M (1 + w) |L|^2 d\mu.$$

By Hölder, (3.8), and Kato's inequality,

$$\int_M w |L|^2 d\mu \leq \|w\|_2 \|L\|_4^2 \leq S_2 \|w\|_2 \int_M (|\nabla L|^2 + |L|^2) d\mu.$$

The coefficient tends to zero uniformly in time, by (3.9). Absorb the gradient term for large j and apply Gronwall's inequality. The initial norm tends to zero by (2.6), proving (4.1). $\square$

We next measure failure of nonnegative sectional curvature by a scalar function of the curvature tensor. On the Euclidean space of algebraic curvature tensors on $\mathbb{R}^4$, define

$$F(B) = \max \left\{ 0, \sup_{\substack{|X|=|Y|=1 \\ \langle X, Y \rangle = 0}} -B(X, Y, X, Y) \right\}. \quad (4.3)$$

Thus $F(\text{Rm})$ is the magnitude of the most negative sectional curvature, or zero if all sectional curvatures are nonnegative.

Proposition 4.2. *There is a uniform constant C such that*

$$\int_{M_j} F(\text{Rm}_{g_j(t)}) d\mu_{g_j(t)} \leq Ct, \quad 0 \leq t < T. \quad (4.4)$$

Proof. The function F is nonnegative, convex, orthogonally invariant, and Lipschitz: it is a supremum of a bounded family of linear functionals and the zero function. Let F_δ be its convolution with a nonnegative smooth radial mollifier supported in a ball of radius δ in the algebraic-curvature tensor space. Then F_δ is smooth, nonnegative, convex, and orthogonally invariant, with

$$|DF_\delta| \leq C, \quad |F_\delta - F| \leq C\delta.$$

Orthogonal invariance makes this construction independent of the orthonormal frame and compatible with the metric connection.

Apply the chain rule to (3.5) in moving orthonormal frames. Convexity gives a favorable Hessian term:

$$\begin{aligned} (\partial_t - \Delta)F_\delta(\text{Rm}) &= DF_\delta(\text{Rm})[\mathcal{Q}(\text{Rm})] - \sum_a D^2F_\delta(\text{Rm})[\nabla_a \text{Rm}, \nabla_a \text{Rm}] \\ &\leq C|\text{Rm}|^2. \end{aligned}$$

After integration, the volume variation contributes $-\int RF_\delta(\text{Rm})d\mu \leq 0$. Closedness and (3.10) therefore imply

$$\int_M F_\delta(\text{Rm}(t))d\mu \leq \int_M F_\delta(\text{Rm}(0))d\mu_{g_j} + Ct \leq C\delta V_0 + Ct.$$

The last step uses $F(\text{Rm}(0)) = 0$. Letting $\delta \downarrow 0$ proves (4.4). $\square$

5 The compact Einstein limit and the gap

We first extract the geometric conclusion that does not use Classification Assumption 1.1.

Proposition 5.1. *For any sequence (2.1), normalize the metrics as in Proposition 2.1 and discard finitely many terms. There are a subsequence, a connected simply connected closed smooth four-manifold N , a metric h on N , and diffeomorphisms $\Phi_j : N \rightarrow M_j$ such that*

$$\text{Ric}_h = 3h, \quad \text{sec}_h \geq 0,$$

and

$$\Phi_j^* g_j(t) \longrightarrow (1 - 6t)h \tag{5.1}$$

smoothly on $N \times I$ for every compact interval $I \subset (0, T)$.

Proof. Fix $t_* = T/2$. By (3.13), the flows have uniform global curvature bounds on every compact time interval in $(0, T)$. At t_* , the local volume lower bound (3.12) and the total volume upper bound V_0 also give a diameter bound. Indeed, points spaced a fixed distance apart along a minimizing geodesic have pairwise disjoint balls of a fixed smaller radius; each such ball has uniformly positive volume. Their number is bounded by V_0 . Write the resulting bound as $\text{diam}(M_j, g_j(t_*)) \leq D_*$.

Absolute sectional curvature bounds and the ball-volume lower bound give a uniform positive injectivity radius at t_* , by the injectivity radius estimate of Cheeger–Gromov–Taylor [4, Theorem 4.7]. Hamilton’s compactness theorem [8] now gives, after passing to a subsequence, smooth pointed convergence to a connected Ricci flow $(N, h(t))$ on $(0, T)$, with $h(t_*)$ complete. One can apply the theorem on nested compact time intervals containing t_* and take a diagonal subsequence. The curvature bounds used here are global in space; hence the complete limit version of compactness applies, without the completeness issue that can arise from purely local bounds [16, Theorem 1.1 and subsequent discussion]. The convergence maps are time-independent embeddings defined on an exhaustion of N .

We explain why the limit is compact and the embeddings are eventually diffeomorphisms. If the complete connected slice $(N, h(t_*))$ were noncompact, it would contain a compact closed ball $\overline{B}(p, L)$ with a point q at distance L , for any $L > 0$. Choose $L > 2D_*$. For large j , a convergence embedding is defined on a neighborhood of this ball, and its pulled-back metric bounds lengths on the ball from below by half their $h(t_*)$ values. Any curve in M_j joining the images of p and q

must have length at least $L/2$: if it leaves the image of the ball, its initial portion must first reach that image's boundary and already has this length. If it stays inside, the same bound follows by pullback. This contradicts the diameter bound. Thus N is compact. Eventually the convergence embeddings are defined on all of N ; their images are open, by the inverse function theorem, and closed, by compactness. Since M_j is connected, each is a diffeomorphism. It follows also that N is simply connected.

Smooth convergence on this compact manifold and (4.1) give

$$\text{Ric}_{h(t)} = \lambda(t)h(t), \quad 0 < t < T.$$

The Ricci-flow equation therefore integrates to

$$h(t) = (1 - 6t)h, \quad h = \frac{1}{1 - 6t_*}h(t_*), \quad \text{Ric}_h = 3h.$$

For each fixed $t > 0$, (4.4) also passes to the limit. The function F is positively homogeneous of degree one. Under metric scaling by a positive constant c , its value on curvature scales by c^{-1} and the four-dimensional volume form scales by c^2 . Consequently

$$(1 - 6t) \int_N F(\text{Rm}_h) d\mu_h \leq Ct, \quad 0 < t < T. \quad (5.2)$$

Let $t \downarrow 0$. The nonnegative continuous integrand must vanish, so $\text{sec}_h \geq 0$. This proves the proposition. $\square$

Proof of Theorem 1.2. If no universal positive gap existed, then for every positive integer j there would be a connected simply connected closed four-manifold (M_j, g_j) with $\text{sec}_{g_j} \geq 0$ and $\mathcal{D}(g_j) < 1/j$, whose smooth manifold is outside the three stated diffeomorphism types. Proposition 5.1 gives a subsequence diffeomorphic to a closed simply connected (N, h) with $\text{Ric}_h = 3h$ and $\text{sec}_h \geq 0$. Classification Assumption 1.1 applies exactly to this metric. Since N is already simply connected, it is one of the three model manifolds, a contradiction. $\square$

Remark 5.2. Proposition 5.1 is independent of the classification input. It asserts smooth convergence only after positive-time regularization. In particular, the recovery of $\text{sec}_h \geq 0$ in (5.2) requires neither convergence of the original metrics at time zero nor preservation of nonnegative sectional curvature by Ricci flow.

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